Statistical analyses

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Abstract

The following presents the data from the different statistical analyses conducted in the study on dislexya across a transparent native language (Spanish L1) and an opaque foreign language (English as an Additional Language [EAL]) within a bilingual context.

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1 Participants

Table 1: Participant Characteristics

Participant	Age	Course	Sex	Dyslexia
1	14	2	1	No
2	14	2	1	No
3	14	2	1	No
4	14	2	1	No
5	14	2	1	Yes
6	14	2	1	Yes
7	14	2	1	Yes
8	14	2	0	Yes
9	14	2	0	Yes
10	14	2	0	Yes
11	14	2	1	Yes
12	14	2	1	Yes
13	14	2	0	Yes
14	14	2	0	Yes
15	14	2	1	No
16	14	2	0	No
17	14	2	1	No
18	14	2	1	Yes
19	14	2	1	Yes
20	14	2	0	No
21	14	2	1	Yes
22	14	2	1	No
23	14	2	1	No
24	14	2	1	Yes
25	14	2	0	Yes
26	14	2	1	No
27	14	2	1	No
28	14	2	0	No
29	14	2	0	No
30	14	2	0	No
31	14	2	0	Yes
32	14	2	1	Yes
33	14	2	1	No
34	14	2	1	Yes
35	14	2	1	No
36	14	2	1	No
37	14	2	0	No
38	14	2	0	No
39	14	2	1	No
40	14	2	1	No
41	14	2	1	No
42	14	2	0	No
43	14	2	0	No
44	14	2	0	No
45	14	2	1	No
46	14	2	1	No
47	14	2	0	No
48	14	2	0	No
49	14	2	0	No
50	14	2	0	No
51	14	2	0	No
52	14	2	0	No
53	14	2	0	No
54	14	2	1	No
55	14	2	0	No
56	14	2	0	No
57	14	2	1	No
58	14	2	1	No
59	14	2	0	No

32 male participants (54.2%), of whom 11 had dyslexia; 27 female participants (45.8%), of whom 7 had dyslexia

2 Spanish tasks

2.1 Descriptive statistics

Table 2: Descriptive Statistics

	Line		Gaps		Correct word		Dictation	
	0	1	0	1	0	1	0	1
Valid	41	18	41	18	41	18	41	18
Missing	0	0	0	0	0	0	0	0
Median	0.000	0.000	5.000	6.000	3.000	4.000	3.000	6.000
Mean	0.610	1.389	5.537	6.222	3.488	4.611	3.854	5.944
Std. Deviation	1.358	2.146	2.740	3.335	1.951	2.429	2.971	3.244
Shapiro-Wilk	0.518	0.639	0.862	0.911	0.905	0.812	0.563	0.975
P-value of Shapiro-Wilk	2.026×10^{-10}	1.699×10^{-5}	1.539×10^{-4}	0.091	0.002	0.002	7.579×10^{-10}	0.890
Minimum	0.000	0.000	1.000	2.000	0.000	2.000	1.000	0.000
Maximum	5.000	5.000	12.000	12.000	10.000	10.000	20.000	12.000

2.2 Independent Samples Mann–Whitney U Test

Table 3: Independent Samples T-Test

					95% CI for Rank-Biserial Correlation	
	W	p	Rank-Biserial Correlation	SE Rank-Biserial Correlation	Lower	Upper
Line	313.000	0.232	−0.152	0.163	−0.443	0.168
Dictation	186.000	0.002	−0.496	0.163	−0.700	−0.218
Gaps	323.000	0.449	−0.125	0.163	−0.420	0.195
Correct word	274.000	0.110	−0.257	0.163	−0.527	0.059

Note. For the Mann–Whitney test, effect size (Hedges' g) is given by the rank biserial correlation.

2.2.1 Assumption checks

Table 4: Test of Equality of Variances (Levene's)

	F	df_1	df_2	p
Line	10.864	1	57	0.002
Dictation	2.178	1	57	0.145
Gaps	1.934	1	57	0.170
Correct word	1.526	1	57	0.222

2.2.2 Raincloud plots

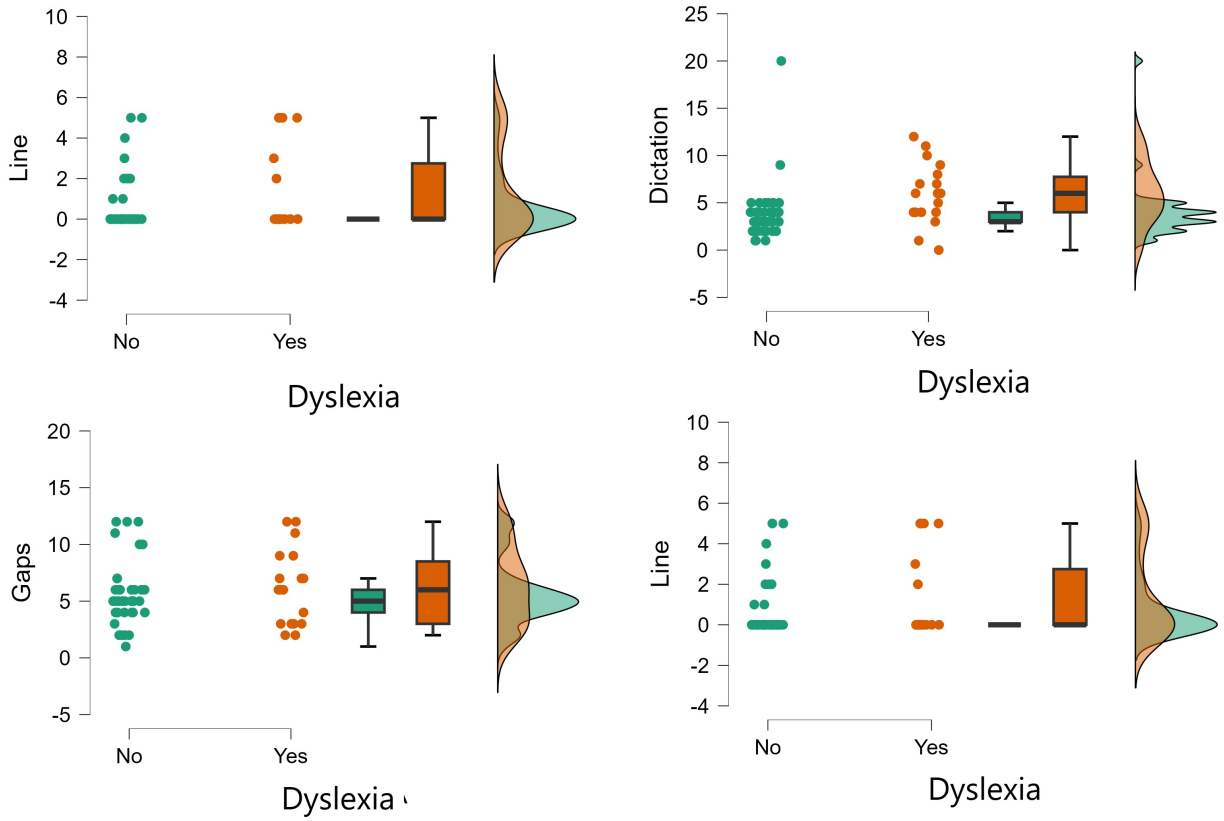


Figure 1: Participants' pattern distribution in Spanish tasks

2.3 Lines task in Spanish

Table 5: Independent Samples T-Test

					95% CI for Rank-Biserial Correlation	
	W	p	Rank-Biserial Correlation	SE Rank-Biserial Correlation	Lower	Upper
Line PR	320.500	0.242	−0.131	0.163	−0.426	0.188
Line GR	291.000	0.059	−0.211	0.163	−0.491	0.108
Line FL	305.000	0.046	−0.173	0.163	−0.460	0.146
Line NSC	300.000	0.106	−0.187	0.163	−0.471	0.133
Line X	332.000	0.335	−0.100	0.163	−0.400	0.218

Note. For the Mann–Whitney test, effect size (Hedges' g) is given by the rank biserial correlation.

2.4 Gaps task in Spanish

Table 6: Independent Samples T-Test

					95% CI for Rank-Biserial Correlation	
	W	p	Rank-Biserial Correlation	SE Rank-Biserial Correlation	Lower	Upper
Gap R vision	364.500	0.939	−0.012	0.163	−0.323	0.301
Gap SC inscribir	417.000	0.361	0.130	0.163	−0.190	0.425
Gap GL regla	343.500	0.463	−0.069	0.163	−0.373	0.248
Gap BL blanco	342.500	0.602	−0.072	0.163	−0.375	0.246
Gap BJ objeto	369.500	1.000	0.001	0.163	−0.311	0.313
Gap DH adhesión	367.500	0.980	−0.004	0.163	−0.316	0.308
Gap D laxitud	382.500	0.805	0.037	0.163	−0.278	0.345
Gap RR aterrizar	319.500	0.343	−0.134	0.163	−0.428	0.186
Gap J injerir	379.000	0.823	0.027	0.163	−0.287	0.336
Gap Y proyecto	283.500	0.088	−0.232	0.163	−0.507	0.086
Gap PS psicología	285.000	0.112	−0.228	0.163	−0.504	0.091
Gap B probación	341.500	0.605	−0.075	0.163	−0.378	0.243

Note. For the Mann–Whitney test, effect size (Hedges' g) is given by the rank biserial correlation.

2.5 Correct word task in Spanish

Table 7: Independent Samples T-Test

					95% CI for Rank-Biserial Correlation	
	W	p	Rank-Biserial Correlation	SE Rank-Biserial Correlation	Lower	Upper
RR ahorrar	323.000	0.207	−0.125	0.163	−0.420	0.195
V convivencia	356.500	0.793	−0.034	0.163	−0.342	0.281
vHv ahuyentar	399.500	0.492	0.083	0.163	−0.235	0.385
S disuasión	302.000	0.137	−0.182	0.163	−0.467	0.138
GU álguido	305.500	0.231	−0.172	0.163	−0.459	0.148
DR padrino	346.000	0.395	−0.062	0.163	−0.367	0.255
PR procesar	332.000	0.335	−0.100	0.163	−0.400	0.218
BR subrayar	351.500	0.736	−0.047	0.163	−0.354	0.268
MN omnívoro	300.000	0.106	−0.187	0.163	−0.471	0.133
Z proeza	268.500	0.047	−0.272	0.163	−0.539	0.043

Note. For the Mann–Whitney test, effect size (Hedges' g) is given by the rank biserial correlation.

2.6 Dictation task in Spanish

Table 8: Independent Samples T-Test

					95% CI for Rank-Biserial Correlation	
	W	p	Rank-Biserial Correlation	SE Rank-Biserial Correlation	Lower	Upper
XH exhaustivo	331.500	0.409	−0.102	0.163	−0.401	0.217
CR escribir	NaN					
NV-V convivencia	279.500	0.041	−0.243	0.163	−0.515	0.075
J-G jengibre	422.500	0.266	0.145	0.163	−0.175	0.437
T-D voluntad	209.000	4.995×10^{-4}	−0.434	0.163	−0.657	−0.141
GR diagrama	NaN					
SC adolescente	277.000	0.050	−0.249	0.163	−0.521	0.068
CL esclavo	341.000	0.486	−0.076	0.163	−0.379	0.242
TL atleta	346.000	0.395	−0.062	0.163	−0.367	0.255
G gesticular	338.500	0.492	−0.083	0.163	−0.385	0.235
CR idiosincrasia	270.500	0.021	−0.267	0.163	−0.534	0.049
vHv desahuciar	338.000	0.440	−0.084	0.163	−0.386	0.234
CL clima	357.500	0.564	−0.031	0.163	−0.340	0.283
FR frágil	337.000	0.173	−0.087	0.163	−0.388	0.231
TR contratiempo	NaN					
CU cuadrilátero	233.500	0.007	−0.367	0.163	−0.609	−0.062
vCv habitación	302.500	0.081	−0.180	0.163	−0.466	0.140
FL flor	357.500	0.564	−0.031	0.163	−0.340	0.283
BR abrigo	357.500	0.564	−0.031	0.163	−0.340	0.283

Note. For the Mann–Whitney test, effect size (Hedges' g) is given by the rank biserial correlation.

3 English tasks

3.1 Descriptive statistics

Table 9: Descriptive Statistics

	Line		Gaps		Dictation		Correct Word	
	0	1	0	1	0	1	0	1
Valid	41	18	41	18	41	18	41	18
Missing	0	0	0	0	0	0	0	0
Median	0.000	2.000	3.000	4.500	5.000	9.500	1.000	3.500
Mean	1.195	2.611	3.780	5.222	5.902	10.444	2.000	3.722
Std. Deviation	1.792	2.227	2.515	3.116	4.364	5.893	2.356	2.585
Shapiro-Wilk	0.698	0.866	0.923	0.927	0.860	0.924	0.755	0.893
P-value of Shapiro-Wilk	6.899×10^{-8}	0.015	0.008	0.171	1.329×10^{-4}	0.150	7.049×10^{-7}	0.044
Minimum	0.000	0.000	0.000	1.000	0.000	2.000	0.000	1.000
Maximum	6.000	6.000	12.000	12.000	20.000	19.000	10.000	10.000

3.2 Independent Samples Mann–Whitney U Test

Table 10: Independent Samples T-Test

					95% CI for Rank-Biserial Correlation	
	W	p	Rank-Biserial Correlation	SE Rank-Biserial Correlation	Lower	Upper
Line	228.000	0.013	−0.382	0.163	−0.620	−0.080
Gaps	277.000	0.128	−0.249	0.163	−0.521	0.068
Dictation	203.000	0.006	−0.450	0.163	−0.668	−0.160
Correct Word	200.000	0.005	−0.458	0.163	−0.674	−0.170

Note. For the Mann–Whitney test, effect size (Hedges' g) is given by the rank biserial correlation.

3.2.1 Assumption checks

Table 11: Test of Equality of Variances (Levene's)

	F	df_1	df_2	p
Line	2.025	1	57	0.160
Gaps	2.512	1	57	0.119
Dictation	5.374	1	57	0.024
Correct Word	0.924	1	57	0.340

3.2.2 Raincloud plots

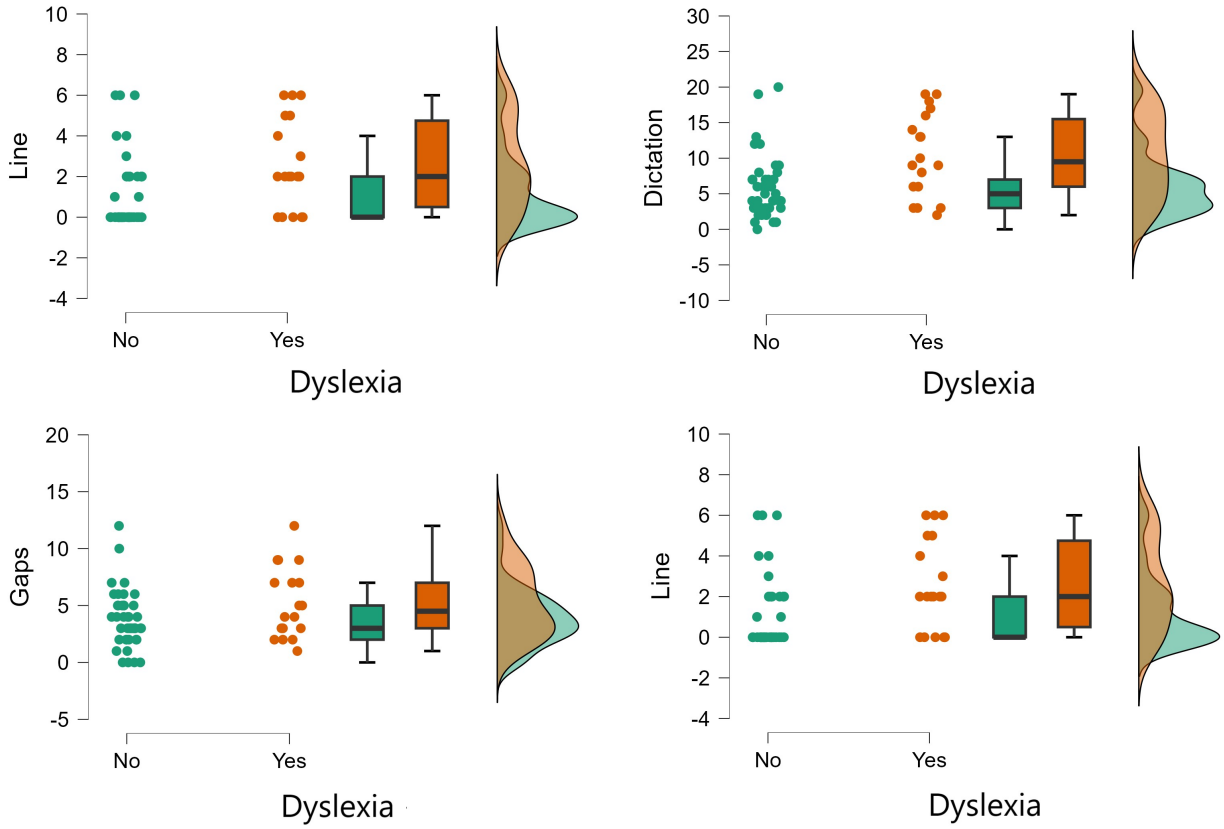


Figure 2: Participants' pattern distribution in English tasks

3.3 Lines task in English

Table 12: Independent Samples T-Test

					95% CI for Rank-Biserial Correlation	
	W	p	Rank-Biserial Correlation	SE Rank-Biserial Correlation	Lower	Upper
Line CK	292.000	0.121	−0.209	0.163	−0.489	0.110
Line CH	220.000	0.006	−0.404	0.163	−0.636	−0.105

Note. For the Mann–Whitney test, effect size (Hedges' g) is given by the rank biserial correlation.

3.4 Gaps task in English

Table 13: Independent Samples T-Test

					95% CI for Rank-Biserial Correlation	
	W	p	Rank-Biserial Correlation	SE Rank-Biserial Correlation	Lower	Upper
Gap TH	336.000	0.487	−0.089	0.163	−0.390	0.229
Gap PH	337.500	0.553	−0.085	0.163	−0.387	0.233
Gap H-	325.500	0.142	−0.118	0.163	−0.415	0.201
Gap BB	407.000	0.461	0.103	0.163	−0.216	0.402
Gap -G	388.000	0.652	0.051	0.163	−0.265	0.358
Gap TR	337.000	0.173	−0.087	0.163	−0.388	0.231
Gap cHv	287.500	0.122	−0.221	0.163	−0.498	0.098
Gap SH	239.000	0.010	−0.352	0.163	−0.598	−0.045
Gap L	357.500	0.564	−0.031	0.163	−0.340	0.283
Gap TT	285.000	0.112	−0.228	0.163	−0.504	0.091
Gap PL	302.500	0.081	−0.180	0.163	−0.466	0.140
Gap -Y	293.500	0.037	−0.205	0.163	−0.486	0.115

Note. For the Mann–Whitney test, effect size (Hedges' g) is given by the rank biserial correlation.

3.5 Correct word task in English

Table 14: Independent Samples T-Test

					95% CI for Rank-Biserial Correlation	
	W	p	Rank-Biserial Correlation	SE Rank-Biserial Correlation	Lower	Upper
CR	277.000	0.050	−0.249	0.163	−0.521	0.068
TTL	283.500	0.088	−0.232	0.163	−0.507	0.086
DR	334.500	0.285	−0.093	0.163	−0.394	0.225
SS	325.500	0.142	−0.118	0.163	−0.415	0.201
PH	341.000	0.486	−0.076	0.163	−0.379	0.242
GHT	334.500	0.285	−0.093	0.163	−0.394	0.225
LL	391.500	0.675	0.061	0.163	−0.256	0.366
CK	320.500	0.242	−0.131	0.163	−0.426	0.188
TH (Thursday)	359.000	0.823	−0.027	0.163	−0.336	0.287
RSD (Thursday)	370.500	0.980	0.004	0.163	−0.308	0.316

Note. For the Mann–Whitney test, effect size (Hedges' g) is given by the rank biserial correlation.

3.6 Dictation task in English

Table 15: Independent Samples T-Test

	W	p	Rank-Biserial Correlation	SE Rank-Biserial Correlation	95% CI for Rank-Biserial Correlation	
					Lower	Upper
-GH high	336.500	0.521	−0.088	0.163	−0.389	0.230
RD bird	224.500	0.004	−0.392	0.163	−0.627	−0.091
LD old	364.000	0.887	−0.014	0.163	−0.324	0.300
TW two	366.500	0.931	−0.007	0.163	−0.318	0.306
TH birthday	254.000	0.022	−0.312	0.163	−0.568	2.771×10^{-4}
CH teacher	270.500	0.021	−0.267	0.163	−0.534	0.049
TTL little	208.000	0.002	−0.436	0.163	−0.659	−0.144
SC school	305.000	0.046	−0.173	0.163	−0.460	0.146
SH shelf	206.500	8.273×10^{-4}	−0.440	0.163	−0.661	−0.149
PR probably	218.000	0.001	−0.409	0.163	−0.639	−0.112
TCH catch	213.000	0.002	−0.423	0.163	−0.649	−0.128
GHT night	313.000	0.265	−0.152	0.163	−0.443	0.168
GHT weight	384.000	0.765	0.041	0.163	−0.275	0.348
TTL battle	236.000	0.006	−0.360	0.163	−0.604	−0.055
RY every	261.500	0.009	−0.291	0.163	−0.553	0.023
TH thief	275.000	0.050	−0.255	0.163	−0.525	0.062
PH microphone	145.000	9.164×10^{-6}	−0.607	0.163	−0.773	−0.364
THR three	288.500	0.074	−0.218	0.163	−0.496	0.101
W witch	398.000	0.579	0.079	0.163	−0.239	0.381
WH when	268.000	0.028	−0.274	0.163	−0.540	0.042

Note. For the Mann–Whitney test, effect size (Hedges' g) is given by the rank biserial correlation.